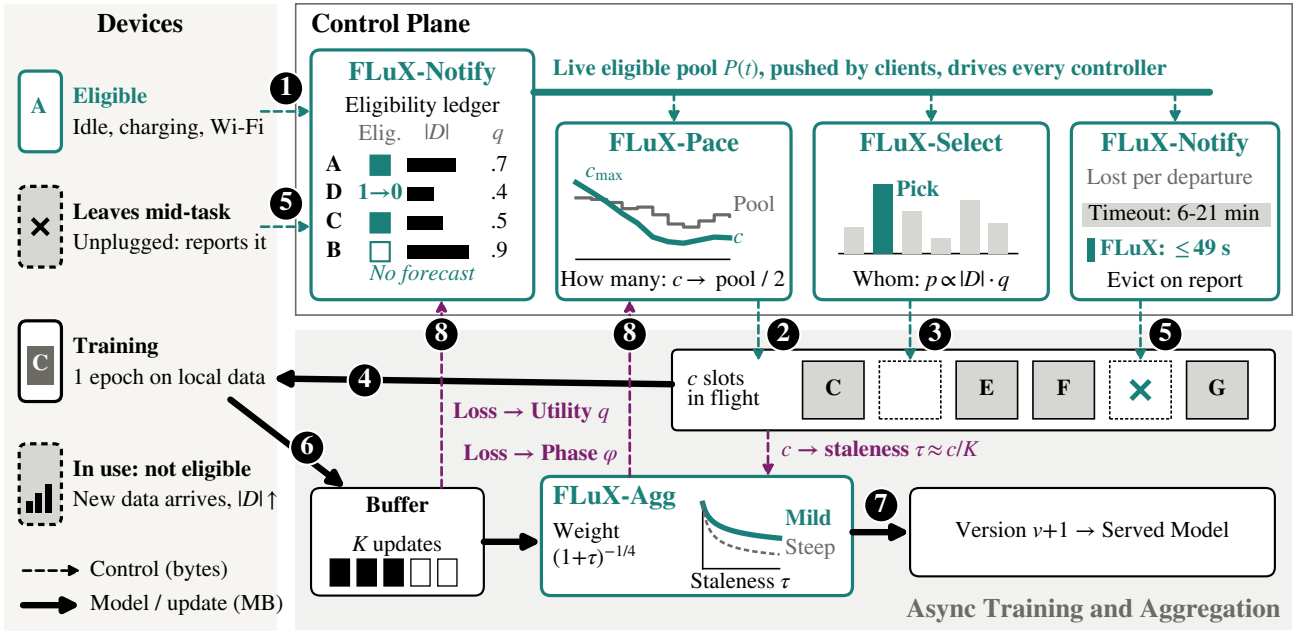
Devices
A
Eligible
Idle, charging, Wi-Fi
Leaves mid-task
Unplugged: reports it
C
Training
1 epoch on local data
In use: not eligible
New data arrives, |D| ↑
Control (bytes)
Model / update (MB)
Control Plane
FLuX-Notify
Eligibility ledger
Elig. |D| q
A .7
D 1→0 .4
C .5
B .9
No forecast
Live eligible pool P(t), pushed by clients, drives every controller
FLuX-Pace
c_max
Pool
c
How many: c → pool / 2
FLuX-Select
Pick
Whom: p ∝ |D| · q
FLuX-Notify
Lost per departure
Timeout: 6-21 min
FLuX: ≤ 49 s
Evict on report
c slots in flight
C
E
F
G
Loss → Utility q
Loss → Phase φ
c → staleness τ ≈ c/K
Buffer
K updates
FLuX-Agg
Weight (1+τ)^−1/4
Mild
Steep
Staleness τ
Version v+1 → Served Model
Async Training and Aggregation